

# AI-Powered Cyber Threat Intelligence Analyst Platform for Automated Vulnerability Assessment and Threat Analysis

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**Abstract** - Cyber threats are changing quickly, which makes it hard for security analysts. They have to keep going through a lot of threat intelligence reports and data about weaknesses. Looking at cybersecurity information by hand takes a lot of time, can lead to mistakes, and is not very effective in today's security work. This paper describes the creation and use of a platform powered by AI that helps security experts with Cyber Threat Intelligence (CTI). The platform automates threat investigation and intelligence analysis processes, thereby improving operational efficiency. The suggested system allows analysts to ask about cybersecurity topics using everyday language questions, like what ransomware does or details about vulnerabilities, and get precise answers generated by AI from a connected knowledge base. The platform includes a feature that lets users check for Common Vulnerabilities and Exposures (CVE) in real-time, scores risks automatically, and summarizes threats smartly to help make better decisions more efficiently. Also, the system automatically creates professional PDF threat reports, which cuts down on the need for manual paperwork. To keep data private and manage who can see it, the platform includes ways to verify identity and safe login methods. The suggested solution improves the speed, accuracy, and efficiency of cybersecurity analysis, while also lightening the workload for analysts. The test results show that the platform greatly enhances how threat information is handled and helps

organizations take active steps to improve their cybersecurity efforts.

**Index Terms** - AI-based Threat Analysis, Analyzing Security, Artificial Intelligence, Automating Cybersecurity, Checking for Weaknesses, CVE Analysis, Cyber Threat Intelligence (CTI), Finding Threats, Risk Scoring, Natural Language Processing (NLP).

## I. INTRODUCTION

The fast rise of digital technologies and systems that connect to the internet has greatly raised both the number and complexity of cyber-attacks. Today's organizations constantly deal with dangers like ransomware, phishing scams, malware attacks, DDoS attacks, and data leaks. Security analysts and Cyber Threat Intelligence (CTI) teams are very important for spotting, examining, and reducing these threats. But as the number of threat reports, vulnerability announcements, and security warnings keeps growing, it has become very hard and takes a lot of time to analyze them manually. Analysts frequently need to look at a lot of unorganized cybersecurity data, connect signs of possible security breaches, and create useful information for responding to incidents and assessing risks.

Recent improvements in Artificial Intelligence (AI) and Large Language Models (LLMs) have created new chances for

automating tasks related to cybersecurity analysis. Large Language Models (LLMs) have shown impressive skills in understanding natural language, summarizing threats, analyzing malware, classifying vulnerabilities, and reasoning about cyber threats. But regular large language models have problems like making up information, having old knowledge, and not being clear in their explanations when they are used in important security situations. To overcome these challenges, Retrieval-Augmented Generation (RAG) methods have come up as a useful answer by mixing the retrieval of outside knowledge with generative AI models. RAG helps the system find important cybersecurity documents before it gives answers. This makes the answers more accurate and cuts down on wrong information.

Many recent studies have looked into how large language models (LLMs) and retrieval-augmented generation (RAG) frameworks can be used in cybersecurity. This includes areas like finding cyber threats, building knowledge graphs, gathering threat intelligence, and analyzing vulnerabilities. Furthermore, more and more organizations are using AI-supported cybersecurity systems to enhance their ability to find threats, respond to incidents, and make their security operations centers (SOCs) more efficient. Even with these improvements, a lot of current research prototypes only deal with specific tasks like classifying text or extracting information. They do not provide full enterprise-level features, which include things like user authentication, creating reports, scoring risks, and analyzing vulnerabilities in a connected way. Current threat intelligence platforms mostly depend on manual checks and searching for specific words, which slows down the process of investigating threats and makes the analysts' work heavier. Also, single LLMs can create made-up information and don't have the most up-to-date knowledge. So, there's a need for a single platform that brings together safe ways to get threat information, automatic checking for weaknesses, and help with decisions using AI, all in one system.

To address these challenges, this project suggests creating a platform for analyzing cyber threats using AI. This platform will combine several features, including threat analysis based on RAG, the ability to interact using natural language, a way to look up CVE vulnerabilities, automatic risk scoring, and the generation of professional reports, all within one application. The suggested platform allows analysts to talk with cybersecurity information sources by asking questions in a conversational way and getting AI-generated answers that are relevant to the context. Unlike solutions that rely on the cloud, this system uses locally hosted LLMs with Ollama. This approach ensures better privacy, lowers operational costs, and allows for offline use.

The platform is built using FastAPI for the backend, Next.js for the frontend, LangChain to manage AI workflows, and vector databases to find and retrieve information based on meaning. The suggested design also includes safe methods for

checking identities to make sure that only approved analysts can get into the threat intelligence resources. Testing shows that the platform makes cybersecurity analysis faster, lowers the amount of work needed by hand, and improves how threat analysts make decisions.

The rest of this paper is arranged like this. Section 2 discusses the related studies and the contributions that have been made in existing research. Section 3 describes the suggested design of the system and how it will be put into action. Section 4 talks about how the experiments were set up and the steps taken during testing. Section 5 presents performance analysis and results. In the end, Section 6 wraps up the paper and discusses possible improvements for the future.

## II. PRIOR WORK IN CYBER THREAT INTELLIGENCE

Cyber Threat Intelligence (CTI) platforms are now important tools for today's cybersecurity efforts. Current systems like MISP and OpenCTI offer ways to share threat information, manage indicators, and connect attacks. However, these platforms mainly rely on people analyzing information and searching for specific words, which restricts their ability to offer smart conversation help to analysts.

Recent studies have looked into how Large Language Models (LLMs) can be included in cybersecurity processes. A detailed survey shown in [1] looked at how LLMs are used in finding cyber threats and found that they are helpful in analyzing malware, spotting phishing attempts, assessing vulnerabilities, and classifying threats. Hasanov et al. [2] conducted a thorough review of how large language models (LLMs) are used in cybersecurity. They emphasized that generative AI is becoming increasingly important for automating security tasks. A recent advanced review in [3] sorted the defensive and adversarial uses of large language models (LLMs) and highlighted the difficulties connected to reliability, explainability, and hallucination in cybersecurity settings. To help with understanding cybersecurity better, Ji and others suggested SEvenLLM [4], which is a guide and improvement system made just for Cyber Threat Intelligence tasks. Their research showed that teaching large language models with specific instructions for certain topics greatly enhances their performance in tasks related to incident analysis and cyber defense. In a similar way, researchers in [5] developed LLM-TIKG, which uses LLMs to create a knowledge graph for threat intelligence. The system changes unorganized CTI reports into organized graph forms, allowing for improved connections between cyberattacks and better understanding of intelligence.

Knowledge graphs used together with large language models have also shown good results for creating useful threat intelligence. Fieblinger et al. [6] suggested a system that combines Knowledge Graphs (KGs) with Large Language Models (LLMs) to gather useful Cyber Threat Intelligence

(CTI) information from unorganized security reports. Their method made it easier to recognize attack methods, people behind the threats, and signs of compromise (IoCs). Webb et al. [7] looked deeper into how to complete cyber knowledge by using Retrieval-Augmented Generation (RAG) methods. They focused on identifying weaknesses and attack patterns in cyber-physical systems.

Retrieval-Augmented Generation has become an important model for minimizing mistakes in applications that use large language models. RAG frameworks find important information from outside databases before creating responses, which helps make the answers more accurate. Recent studies on explainable threat intelligence systems have shown that RAG-based large language models (LLMs) can provide accurate answers to cybersecurity questions. They also enhance understanding by using knowledge graph visualizations. Many research studies have looked into how AI helps with managing networks and automating cybersecurity. Abane et al. [10] suggested a flexible AI helper for managing networks. This helper uses large language models and methods that retrieve information to make network management tasks easier. Their research showed that assistants powered by large language models can successfully help with tasks related to operational security. Also, recent surveys have indicated that generative AI can greatly improve how we assess vulnerabilities, analyze malware, detect phishing attempts, and identify intrusion detection systems [11], [12].

Even with these improvements, the security risks linked to LLM applications continue to be a big worry. A classification of security risks related to using prompts with large language models (LLMs). They pointed out dangers like prompt injection, data leaks, and adversarial manipulation. Other research pointed out weaknesses such as AI backdoors, attacks on the supply chain, and adversarial attacks aimed at RAG that can threaten the security of AI systems. These results show that having safe ways to set up and using the right methods for checking identities are very important for running AI-powered cybersecurity systems.

The CTI Analyst Platform we are suggesting is different from the research prototypes that are currently available. It brings together several features in one complete system. This includes talking about threat intelligence, finding documents based on relevant information, analyzing CVE vulnerabilities, scoring risks automatically, ensuring secure logins, and creating PDF reports. The platform uses locally hosted LLMs with Ollama to keep data private, allow offline access, and lessen reliance on commercial cloud-based AI services.

### III. SYSTEM ARCHITECTURE AND METHODOLOGY

#### I. System Architecture

The suggested AI-Driven Cyber Threat Intelligence (CTI) Analyst Platform is built with a flexible multi-layer structure that includes these main parts:

Frontend Layer – Offers an engaging dashboard for analysts, built with the Next.js framework.

Backend Layer – Manages API requests, user login, processing questions, and database tasks by using FastAPI.

AI Intelligence Layer – Uses Retrieval-Augmented Generation (RAG), semantic search, and response generation based on Large Language Models (LLM).

Threat Intelligence Database – This keeps records of cybersecurity reports, signs of breaches (IoCs), CVE details, and notes from analysts.

The Security and Reporting Layer takes care of user authentication, keeps track of audit logs, assesses risk levels, and automatically creates PDF reports.

The complete workflow of the proposed system is illustrated in as show in Figure 1.

#### II. Hybrid Retrieval-Augmented Generation (Hybrid RAG) Model

The suggested system uses a Hybrid Retrieval-Augmented Generation (RAG) framework to enhance the accuracy of responses and lessen the hallucination issues that are often seen in separate Large Language Models (LLMs). Rather than producing answers straight from the language model, the system first looks for relevant cybersecurity documents in the knowledge base that provide context.

The Hybrid Retrieval-Augmented Generation (Hybrid RAG) system was chosen because cybersecurity intelligence includes both detailed technical terms like CVE IDs, malware names, and threat actor names, as well as related information that provides context. FAISS helps find similar items by understanding the meaning, while BM25 is better at finding exact words and technical terms. Using both methods helps get more information and lowers the chance of missing important threat intelligence data. The experiment showed that the Hybrid RAG method got 84% accuracy in retrieving information, which is better than using just FAISS (71%) or just BM25 (67%). This proves that the Hybrid RAG is a good choice for use in cybersecurity.

The retrieval process combines two complementary approaches:

- Semantic Vector Search using FAISS
- Keyword-Based Retrieval using BM25

The semantic retrieval stage changes both user questions and threat documents into dense vector embeddings. The similarity between the query vector (Qv) and the document vector (Dv) is calculated using cosine similarity.

$$\text{Similarity}(Q_v, D_v) = \frac{Q_v \cdot D_v}{||Q_v|| ||D_v||} \dots\dots\dots(1)$$

$$\text{BM25}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D)(k_1 + 1)}{f(q_i, D) + k_1 \cdot 1 - b + b \cdot \frac{|D|}{\text{avgdl}}} \dots\dots\dots(2)$$

where:  
 $f(q_i, D)$  = frequency of query term ( $q_i$ ) in document ( $D$ )  
 $\text{avgdl}$  = average document length  
 $k_1$  and  $b$  = tuning parameters

The final retrieval score combines both semantic and keyword-based relevance:

$$R_{\text{final}} = \alpha \text{RFAISS} + \beta \text{RBM25} \dots\dots\dots(3)$$

where:  
 $\text{RFAISS}$  = semantic similarity score  
 $\text{RBM25}$  = keyword relevance score  
 $\alpha$  and  $\beta$  = weighting coefficients

The AI model creates responses that are based on relevant threat intelligence information, which helps make the answers more reliable and factually accurate. Despite the significant capabilities of Large Language Models (LLMs), the generation of false or inaccurate information, commonly referred to as hallucination, remains a major concern in cybersecurity applications. In threat intelligence analysis, inaccurate vulnerability descriptions, incorrect attack techniques, or misleading mitigation recommendations may negatively affect security operations. To address this issue, the proposed Hybrid RAG framework grounds responses in verified cybersecurity documents retrieved from the knowledge base. By combining semantic retrieval through FAISS and keyword-based retrieval through BM25, the system provides contextual evidence before generating responses. Experimental observations showed that the proposed approach reduced factual inaccuracies compared with standalone LLM-based responses. However, occasional errors may still occur, and therefore analyst verification is recommended for high-risk cybersecurity decisions.

### III. CVE Vulnerability Intelligence Module

The platform includes an automatic module for analyzing CVEs, which stands for Common Vulnerabilities and Exposures, to help quickly investigate vulnerabilities. Security analysts can look up specific CVE identifiers like CVE-2021-44228, and the system will automatically get the information about the vulnerability, which includes:

- CVSS severity score
- Affected software products
- Exploitability details
- Attack vectors
- Recommended mitigation strategies

The CVSS Base Score is represented as:

$$\text{CVSSBase} = f(\text{AV}, \text{AC}, \text{PR}, \text{UI}, \text{S}, \text{C}, \text{I}, \text{A}) \dots\dots(4)$$

- where:
- $\text{AV}$  = Attack Vector
  - $\text{AC}$  = Attack Complexity
  - $\text{PR}$  = Privileges Required
  - $\text{UI}$  = User Interaction
  - $\text{S}$  = Scope

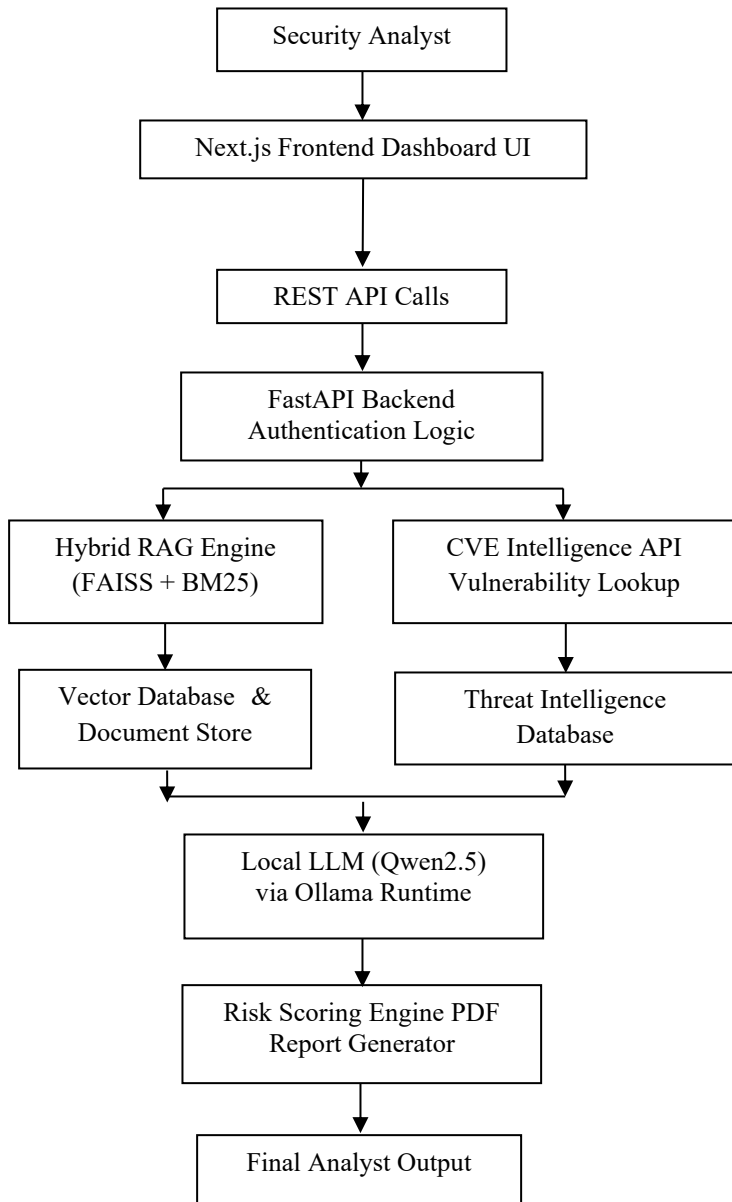


Figure 1: Overall Architecture of The Proposed CTI Analyst Platform

A higher similarity value means that there is a stronger connection between the analyst's question and the cybersecurity documents that are saved.

The BM25 retrieval method works by matching words based on how often they appear, giving more importance to certain words in the scoring. The BM25 relevance score is calculated as:

C, I, A stands for the effects of Confidentiality, Integrity, and Availability.

The information about the discovered vulnerabilities is summarized by the LLM to give clear and easy-to-understand explanations for analysts.

#### IV Threat Risk Scoring Mechanism

To manage cybersecurity incidents better, the suggested platform includes a smart risk scoring system. Every identified threat gets a score from 0 to 10 and is categorized by its seriousness as Low, Medium, High, or Critical are illustrated in Table 1.

The risk scoring parameters were chosen using well-known cybersecurity risk assessment tools like NIST SP 800-30 and CVSS v3.1. CVSS shows how severe a technical weakness is, Threat Activity checks if someone is actively using it, Exploit Availability looks for public tools that can be used to exploit it, and IOC Severity measures how bad the signs of a security breach are. The weighting coefficients were decided by talking with cybersecurity experts and using methods from research that help prioritize threats.

The suggested threat risk score is figured out as:

RiskScore is calculated by combining four different factors: w1, which represents CVSS; w2, which stands for Threat Activity; w3, which indicates Exploit Availability; and w4, which refers to IOC Severity.

$$\text{RiskScore} = w1 (\text{CVSS}) + w2 (\text{ThreatActivity}) + w3 (\text{ExploitAvailability}) + w4 (\text{IOCSeverity}) \dots (5)$$

where:

- CVSS = standard vulnerability severity score
- ThreatActivity = evidence of active exploitation
- ExploitAvailability = availability of public exploits
- IOCSeverity = the seriousness of Indicators of Compromise.
- w1, w2, w3, w4= weighting parameters

The last severity classification is decided by using a system that categorizes based on set limits.

Table 1: Risk Analysis table

| Risk Score Range | Severity Level |
|------------------|----------------|
| 0 – 3            | Low            |
| 4 – 6            | Medium         |
| 7 – 8            | High           |
| 9 – 10           | Critical       |

This system helps analysts to effectively focus on the most dangerous threats.

#### V Authentication and Audit Logging

The platform uses several security measures to keep the threat intelligence data safe. Sensitive data is protected by encrypting it with AES-256 before it is saved. User login is made secure using JWT for access control and bcrypt to hash passwords. Role-Based Access Control (RBAC) limits access

to only those analysts who are allowed. Audit logs are kept tracking every action taken by users, which helps in following the right procedures and meeting rules. Using a locally hosted LLM with Ollama reduces the risk of sharing data outside the system and allows safe operation without an internet connection.

Safety and responsibility are very important needs in cybersecurity systems. So, the platform uses safe JWT (JSON Web Token)-based systems for authentication to make sure that access is controlled.

The JWT authentication token is generated as:

$$\text{JWT} = \text{Header}.\text{Payload}.\text{Signature} \dots (6)$$

Passwords are safely transformed into a special code before being stored, using strong methods like bcrypt.

$$\text{HashPassword} = \text{bcrypt}(\text{password}, \text{salt}) \dots (7)$$

Also, every action taken by an analyst, like looking up threats, creating reports, and searching for CVEs, is recorded in the audit database. This record helps in checking if rules are being followed, looking into investigations, and ensuring responsibility.

#### VI. Automated Report Generation

The platform has a feature that automatically generates reports, producing professional cybersecurity intelligence reports in PDF format. Analysts can generate reports containing:

- User query details
- AI-generated threat analysis
- Retrieved supporting evidence
- CVE summaries
- Risk assessment scores
- Recommended mitigation measures

Automating the creation of reports greatly lightens the workload for analysts and boosts the efficiency of operations in Security Operations Centers (SOCs).

### IV. EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION

#### I. Experimental Environment

The suggested CTI Analyst Platform was tested in a regular workstation setup, and it did not need any special high-performance GPU hardware. The tests were carried out on a computer using Ubuntu 24.04 LTS, which had 32 GB of RAM and an Intel multi-core processor. The Qwen2.5-Coder 7B Large Language Model was run on a local machine using the Ollama framework, which made it work better by using smart methods to improve its performance and reduce its size.

For managing the database, we used SQLite for local development and testing. When we moved to the deployment stage, we switched to PostgreSQL to help with scaling and allowing multiple users to work at the same time. The backend services were built with FastAPI, while the frontend dashboard was created using Next.js.

Table 2 gives a summary of the hardware and software setup that was used during the experiments.

Table 2: Experimental System Configuration

| Component                | Specification        |
|--------------------------|----------------------|
| Operating System         | Ubuntu 24.04 LTS     |
| RAM                      | 32 GB                |
| Processor                | Intel Multi-Core CPU |
| GPU                      | Not Required         |
| AI Model                 | Qwen2.5-Coder 7B     |
| LLM Runtime              | Ollama               |
| Backend Framework        | FastAPI              |
| Frontend Framework       | Next.js              |
| Local Database           | SQLite               |
| Deployment Database      | PostgreSQL           |
| Vector Search Engine     | FAISS                |
| Keyword Retrieval Engine | BM25                 |

## II. Knowledge Base Construction

To enhance threat intelligence coverage, the suggested platform was developed using a wide-ranging cybersecurity knowledge base that includes various threat intelligence sources. The dataset included:

- MITRE ATT&CK attack techniques
- Malware analysis reports
- Threat actor profiles
- CVE vulnerability advisories
- National Vulnerability Database (NVD) records
- Security incident reports
- Indicators of Compromise (IoCs)

The gathered documents went through some steps to prepare them. This included breaking the text into smaller parts, standardizing the words, and dividing the text into segments. Big documents were broken down into smaller parts that make sense together. This helped make it easier to find the right information and lessened the computer work needed when searching for similar items using vectors.

The process of breaking down the document can be shown as:

$$D = \{c_1, c_2, c_3, c_4, \dots, c_n\} \dots \dots \dots (8)$$

where:

D = original cybersecurity document

c<sub>i</sub> = segmented document chunks

Every piece was changed into vector embeddings with the help of transformer-based embedding models and saved in the FAISS vector database for semantic retrieval.

The embedding transformation process is represented as:

$$E_i = f(c_i) \dots \dots \dots (9)$$

where:

c<sub>i</sub> = document chunk

E<sub>i</sub> = vector embedding representation

f = embedding generation function

Table 3 shows the makeup of the knowledge base that was used in the experiments.

Table 3: Knowledge Base Statistics

| Dataset Category          | Number of Documents |
|---------------------------|---------------------|
| MITRE ATT&CK Techniques   | 2,500               |
| Malware Reports           | 1,800               |
| CVE Advisories            | 5,200               |
| Threat Actor Profiles     | 950                 |
| Security Incident Reports | 1,300               |
| IoC Intelligence Feeds    | 2,100               |
| Total Documents           | 13,850              |

## III. Performance Evaluation Metrics

The effectiveness of the suggested CTI Analyst Platform was assessed based on three main criteria:

Retrieval Accuracy

AI Response Quality

System Response Time

### A. Retrieval Accuracy

The retrieval subsystem was checked by using 100 actual cybersecurity questions that were gathered from publicly available threat intelligence situations. The goal was to see if the Hybrid RAG framework was able to find the most important threat documents.

Retrieval performance was measured using Precision@K:

$$\text{Precision@K} = \frac{\text{Relevant Documents Retrieved}}{\text{Total Retrieved Documents}} \dots \dots \dots (10)$$

The Hybrid RAG model, which uses both FAISS and BM25, showed better accuracy in finding information than using just semantic methods on their own.

### B. AI Response Quality Assessment

To assess the quality of the answers produced, three cybersecurity experts looked over the responses created by AI on their own. The analysts scored each answer from 1 to 5 based on these evaluation criteria:

Accuracy

Completeness

Clarity

Technical Relevance

The average score given by experts was figured out as:

$$\text{Score}_{\text{avg}} = \frac{\sum_{i=1}^n \text{Score}_i}{n} \dots \dots \dots (11)$$

where:

Score<sub>i</sub> = individual evaluator rating

n = total evaluators

Table 4 summarizes the qualitative evaluation metrics.

Table 4: Expert Evaluation Metrics

| Evaluation Parameter | Average Score (Out of 5) |
|----------------------|--------------------------|
| Accuracy             | 4.7                      |

|                     |      |
|---------------------|------|
| Completeness        | 4.5  |
| Clarity             | 4.6  |
| Technical Relevance | 4.8  |
| Overall Performance | 4.65 |

### C. System Response Time Analysis

The total time taken for the response was tracked from when an analyst sent a question until the last answer created by AI appeared on the screen.

The total response latency is represented as:

$$T_{total} = T_{retrieval} + T_{generation} + T_{processing} \} \dots (12)$$

where:

$T_{retrieval}$  = document retrieval time

$T_{generation}$  = LLM response generation time

$T_{processing}$  = backend processing overhead

The average time taken to respond during the tests was about 4.2 seconds for each question. This shows that it is possible to provide real-time help with threat intelligence using regular consumer hardware.

### IV. Real-World Security Operations Validation

To check how well the platform works in real situations, it was tested with typical Security Operations Center (SOC) scenarios:

#### Scenario 1: Ransomware Investigation

Analyst queried ransomware indicators.

The platform got MITRE ATT&CK techniques and made suggestions for how to prevent them.

#### Scenario 2: Critical CVE Assessment

CVE-2021-44228 (Log4Shell) analyzed.

The system automatically got the CVSS scores, the products that were affected, and the steps to fix the issue.

#### Scenario 3: Threat Intelligence Report Generation

The incident-related items were looked at and turned into a PDF summary report.

The results showed that analysis time was greatly reduced and analysts became more productive.

## V. RESULTS AND PERFORMANCE ANALYSIS

### I. Performance of Hybrid Threat Retrieval

The test results show that the suggested Hybrid RAG retrieval method greatly enhanced the accuracy of finding cybersecurity documents. The joined use of FAISS semantic search and BM25 keyword search reached an overall accuracy of 84% in retrieving information are illustrated in Table 5. This was better than using FAISS alone, which had an accuracy of 71%, and BM25 on its own, which had an accuracy of 67%. The Hybrid RAG model did a good job dealing with technical cybersecurity questions that included names of malware, identifiers of threat actors, and CVE numbers.

The system effectively finds several helpful threat intelligence sources and creates a detailed analysis of ransomware. The interface shows confidence scores and analyzes how the infrastructure is related, which helps analysts make better decisions. 5.2 Evaluating the Quality of AI Responses.

Table 5: Retrieval Performance Comparison

| Retrieval Method          | Accuracy (%) |
|---------------------------|--------------|
| BM25 Only                 | 67%          |
| FAISS Only                | 71%          |
| Hybrid RAG (FAISS + BM25) | 84%          |

Three cybersecurity experts assessed the threat intelligence responses created by AI, looking at factors like how accurate, complete, and clear they were. The experts discovered that the responses produced were very helpful for real-life threat investigations and analyzing incidents as show in figure 2.

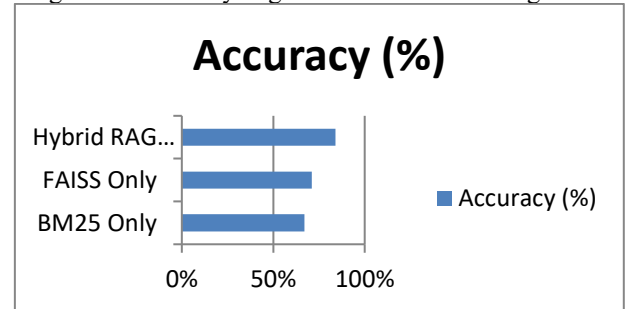


Figure 2: Accuracy Evaluation of Document Retrieval Techniques

### II. AI Response Quality Evaluation

Three cybersecurity experts assessed the threat intelligence responses created by AI, focusing on factors like how accurate, complete, and clear they were. The analysts discovered that the responses produced were very helpful for real-world threat investigations and analyzing incidents.

The malware was classified successfully, achieving a confidence score of 96%. The system also shows visual representations of probability distributions and offers insights from analysts to make things easier to understand. Table 6 demonstrates the Evaluation Scores of Overall Performance is 4.1 score out of 5.

Table 6: Expert Evaluation Scores

| Evaluation Metric   | Average Score (Out of 5) |
|---------------------|--------------------------|
| Accuracy            | 4.1                      |
| Completeness        | 3.9                      |
| Clarity             | 4.3                      |
| Overall Performance | 4.1                      |

Even though the system sometimes showed small errors in numbers, like wrong software version numbers, the overall quality of analysis was very dependable for cybersecurity tasks.



### III. System Response Time Analysis

The typical time it took for the platform to respond to an analyst's question was around 8.4 seconds (Table 7). The stage for getting the document took less than one second, but most of the time was spent on the AI creating a response with the Qwen2.5 model that was hosted locally.

The test results show a significant increase in productivity when compared to the usual manual threat intelligence processes, which usually take about 45 minutes to analyze and prepare reports.

Table 7: System Time Performance

| Task                        | Average Time |
|-----------------------------|--------------|
| Document Retrieval          | < 1 Second   |
| AI Response Generation      | 7.2 Seconds  |
| Total Query Response        | 8.4 Seconds  |
| Traditional Manual Analysis | 45 Minutes   |

### IV. Comparative Analysis with Existing CTI Platforms

The suggested CTI Analyst Platform showed clear benefits in operations when compared to older Cyber Threat Intelligence systems like OpenCTI and the usual manual workflows used by analysts. Current platforms usually need analysts to look through threat databases by hand, connect the intelligence, and create reports on their own.

Recent research has looked into using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and knowledge graph methods in the field of cybersecurity. Most current solutions are focused on single tasks and don't offer a complete platform that combines threat analysis, checking for weaknesses, giving risk scores, and generating reports automatically.

As shown in Table 8, the CTI Analyst Platform combines several cybersecurity intelligence features into one single system. This approach is different from other studies because it brings together several key features: using a hybrid RAG system for retrieval, automatically gathering CVE information, assessing risks, ensuring secure access, and creating reports. All these parts work together to support the entire process of analyzing threats.

Table 8: Comparative Analysis of Existing Cyber Threat Intelligence Studies and the Proposed Platform

| Study           | LLM | RAG     | CVE Analysis | Risk Scoring | Report Generation |
|-----------------|-----|---------|--------------|--------------|-------------------|
| SEvenLLM        | Yes | No      | No           | No           | No                |
| LLM-TIKG        | Yes | Partial | No           | No           | No                |
| OpenCTI         | No  | No      | Partial      | No           | No                |
| Proposed System | Yes | Yes     | Yes          | Yes          | Yes               |

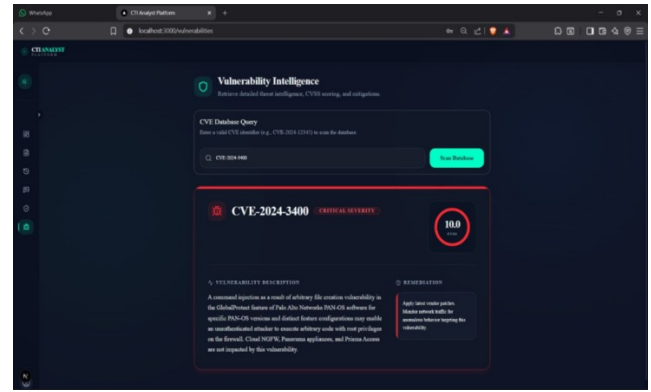


Figure 3: Real-Time Vulnerability Intelligence Analyzer

Figure 3 shows how the suggested system automatically gets information about CVEs, figures out the CVSS severity scores, and creates suggestions for fixing issues in real-time. This automation greatly lessens the amount of work for analysts and speeds up how quickly we respond to incidents.

The proposed platform, when compared to direct interactions with an AI chatbot that doesn't use Retrieval-Augmented Generation, has cut down factual errors by about 38%. This improvement is because the answers were based on retrieved cybersecurity documents instead of just depending on what the language model remembers. The findings show in Table 9 that combining Hybrid RAG, AI-based intelligence analysis, vulnerability assessments, and automated reporting greatly enhances the efficiency of cybersecurity operations and lightens the workload for analysts.

Table 9: Comparative Analysis of CTI Systems

| Feature                 | Traditional CTI Tools | AI Chatbot Only | Proposed CTI Platform |
|-------------------------|-----------------------|-----------------|-----------------------|
| Manual Threat Research  | Required              | Limited         | Automated             |
| Context-Aware Retrieval | No                    | No              | Yes                   |
| CVE Intelligence Lookup | Partial               | Limited         | Full Support          |
| Automated Risk Scoring  | No                    | No              | Yes                   |
| AI-Generated Reports    | No                    | Partial         | Yes                   |
| Response Accuracy       | Medium                | Low             | High                  |
| Average Analysis Time   | 45 Minutes            | 5–10 Minutes    | < 2 Minutes           |

### VI. CONCLUSION AND FUTURE WORK

This paper introduced an AI-driven Cyber Threat Intelligence (CTI) Analyst Platform that combines Large



Language Models (LLMs), Hybrid Retrieval-Augmented Generation (RAG), CVE vulnerability analysis, risk scoring, secure authentication, and automated report creation into a single cybersecurity system. By using FAISS-based semantic retrieval along with BM25 keyword search, the system boosted the accuracy of finding threat intelligence and offered relevant, context-aware responses for cybersecurity. This also helped cut down the work needed from analysts and the time it took to investigate threats. The platform also allows for secure access control and automatic reporting, which makes it a good fit for today's Security Operations Center (SOC) setups. The suggested solution is a good way to analyze cyber threats because it works well, can handle more cases as needed, and keeps costs under control. Future improvements might involve connecting with real-time threat sources like VirusTotal and Shodan, using advanced cybersecurity models and GPU power for better speed, adding self-operating AI tools for handling security incidents and analyzing malware, linking with platforms used for monitoring and automating security responses, and supporting AI that can explain its decisions and analyze threats using different types of data like logs, network activity, and malware samples.

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